

The difference of the first forward differences are called second forward differences. i.e.,

$$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$$

$$\Delta^2 y_1 = \Delta y_2 - \Delta y_1$$

$$\Delta^2 y_{n-2} = \Delta y_{n-1} - \Delta y_{n-2}$$

FORWARD DIFFERENCE TABLE:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	----	$\Delta^n y$
x_0	y_0	Δy_0	$\Delta^2 y_0$	$\Delta^3 y_0$		
x_1	y_1	Δy_1	$\Delta^2 y_1$		----	$\Delta^n y_0$
x_2	y_2	Δy_2				
x_3	y_3					
$\vdots$	$\vdots$					
x_{n-2}	y_{n-2}			$\Delta^3 y_{n-3}$		
x_{n-1}	y_{n-1}	Δy_{n-1}	$\Delta^2 y_{n-2}$			
x_n	y_n					

The first entries in the table Δy_0 , $\Delta^2 y_0$, $\Delta^3 y_0$, ---- $\Delta^n y_0$ are called the leading forward differences.

BACKWARD DIFFERENCE:

Let $f(x)$ be a given function and h be the difference of interval, then the backward difference $\nabla f(x)$ is defined as

$$\nabla f(x) = f(x) - f(x-h)$$

$$\nabla f(x_n) = f(x_n) - f(x_{n-1}) \therefore \nabla y_n = y_n - y_{n-1}$$

$$\nabla f(x_{n-1}) = f(x_{n-1}) - f(x_{n-2}) \therefore \nabla y_{n-1} = y_{n-1} - y_{n-2}$$

$$\nabla y_2 = y_2 - y_1$$

$$\nabla y_1 = y_1 - y_0$$

Similarly second backward difference are defined by ,

$$\nabla^2 y_n = \nabla y_n - \nabla y_{n-1}$$

$$\nabla^2 y_{n-1} = \nabla y_{n-1} - \nabla y_{n-2}$$

$$\nabla^2 y_2 = \nabla y_2 - \nabla y_1$$

BACKWARD DIFFERENCE TABLE :

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	---	$\nabla^n y$
x_0	y_0					
x_1	y_1	∇y_1	$\nabla^2 y_2$	$\nabla^3 y_2$		$\nabla^n y_n$
x_2	y_2	∇y_2				
x_3	y_3					
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$		
$\vdots$	$\vdots$	$\vdots$		$\nabla^3 y_n$		
x_{n-1}	y_{n-1}	∇y_n	$\nabla^2 y_n$			
x_n	y_n					

The last entries in the table ∇y_n , $\nabla^2 y_n$, $\nabla^3 y_n$ --- $\nabla^n y_n$ are called the leading backward differences.